Tutorial 8: Number Theory II

31 December 2023 22:10

Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.

a) $a = 2, m = 17$
b) $a = 34, m = 89$
c) $a = 144, m = 233$
d) $a = 200, m = 1001$

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a) $17 = 8(2) + 1 \Rightarrow 1 = 1(17) - 8(2)$

$2x \equiv_{17} 1 \Rightarrow x \equiv_{17} -8 \equiv_{17} 9$

b)

$89 = 2(34) + 21$
$34 = 1(21) + 13$
$21 = 1(13) + 8$
$13 = 1(8) + 5$
$8 = 1(5) + 3$
$5 = 1(3) + 2$
$3 = 1(2) + 1$
$2 = 2(1) + 0$

s	t
1	0
0	1
1	-2
-1	3
2	-5
-3	8
5	-13
-8	21
13	-34

$s_i = s_{i-2} - q_{i-1} s_{i-1}$

$1 = (13)89 - 34(34)$

$x \equiv_{89} -34 \equiv_{89} 55$

Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m. [*Hint:* Assume that there are two solutions b and c of the congruence $ax \equiv 1 \pmod{m}$. Use Theorem 7 of Section 4.3 to show that $b \equiv c \pmod{m}$.]

Screen clipping taken: 31/12/2023 22:30

$\gcd(a, m) = 1$

Let $x_0 \in \mathbb{Z}$ be such that

$$a x_0 \equiv 1 \pmod{m} \longrightarrow (1)$$

Let x_1 also be a solution such that

$$a x_1 \equiv 1 \pmod{m} \longrightarrow (2)$$

multiply (1) by x_1

$$\Rightarrow \quad x_1 a x_0 \equiv_m x_1$$

Since $\quad x_1 a \equiv_m 1 \quad \Rightarrow \quad x_0 \equiv_m x_1$

$\Rightarrow$ x_0 and x_1 are same solution modulo m.

Show that an inverse of a modulo m, where a is an integer and $m > 2$ is a positive integer, does not exist if $\gcd(a, m) > 1$.

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if a has an inverse x, then

$$ax - 1 = mk \qquad k \in \mathbb{Z}$$

$$\Rightarrow \quad ax - mk = 1$$

let $d = \gcd(a, m)$ since $d \mid a$ and $d \mid m$

let $d = \gcd(a, m)$ since $d|a$ and $d|m$

$\Rightarrow d|1 \Rightarrow \boxed{d=1}$

Thus, if $d > 1$ a solution doesnot exist.

Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 6.

a) $34x \equiv 77 \pmod{89}$

b) $144x \equiv 4 \pmod{233}$

c) $200x \equiv 13 \pmod{1001}$

Screen clipping taken: 31/12/2023 22:41

a) $\gcd(89, 34) = 1$ inverse of 34 is 55

$\Rightarrow x \equiv_{89} 4 \cdot 55 \equiv_{89} 52$

b)

$233 = (1)144 + 89$
$144 = (1)89 + 55$
$89 = (1)55 + 34$
$55 = (1)34 + 21$
$34 = (1)21 + 13$
$21 = (1)13 + 8$
$13 = (1)8 + 5$
$8 = (1)5 + 3$
$5 = (1)3 + 2$
$3 = (1)2 + 1$
$2 = (2)(1) + 0$

	s	t
	1	0
1	0	1
1		-1
1		2
1		-3
1		5
1		-8
1		13
1		-21
1		34
1		-55
1		89

$1 = (89)(144) - (55)233$

$x \equiv 4 \times 89 \equiv 356 \equiv 123$

c) $1001 = 5(200) + 1$

$1 = 1001 - 5(200)$

$x \equiv -65 \equiv 936$

Find the solutions of the congruence $12x^2 + 25x \equiv 10 \pmod{11}$. [*Hint:* Show the congruence is equivalence to the congruence $12x^2 + 25x + 12 \equiv 0 \pmod{11}$. Factor the left-hand side of the congruence; show that a solution of the quadratic congruence is a solution of one of two different linear congruences.]

Screen clipping taken: 31/12/2023 23:00

$12x^2 + 25x \equiv 10 \pmod{11}$

$\Rightarrow 12x^2 + 25x - 10 \equiv 0 \pmod{11}$

$\Rightarrow 12x^2 + 25x + 1 \equiv 0 \pmod{11}$

$\Rightarrow 12x^2 + 25x + 12 \equiv 0 \pmod{11}$

$12x^2 + 25x + 12 = (3x + 4)(4x + 3)$

$\Rightarrow (3x+4)(4x+3) \equiv 0 \pmod{11}$

$\Rightarrow (3x+4)(4x+3) = 11n \quad n \in \mathbb{Z}$

$\Rightarrow$ either, $3x + 4 = 11$ or $4x + 3 = 11$

$\notin \mathbb{Z}$

$x = 2$

$\Rightarrow 12x^2 + 25x = 98 \equiv_{11} 10$ ✓

Show that if m is an integer greater than 1 and $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m/\gcd(c, m)}$.

Screen clipping taken: 31/12/2023 23:13

$m > 1 \in \mathbb{Z}$

$$ac \equiv bc \pmod{m}$$

$\Rightarrow \quad m \mid ac - bc \quad \Rightarrow \quad c(a-b) = nm \quad , n \in \mathbb{Z}$

let $d = \gcd(c, m) \Rightarrow d \mid c, \quad d \mid m$

dividing the equation by d,

$$\frac{c}{d}(a-b) = n\frac{m}{d}$$

$\Rightarrow \gcd\left(\frac{c}{d}, \frac{m}{d}\right) = 1$

since $\frac{m}{d} \mid \left(\frac{c}{d}\right)(a-b)$

$\Rightarrow \frac{m}{d} \mid a-b \Rightarrow \quad a \equiv b \pmod{m/d}$ ✓

Use the construction in the proof of the Chinese remainder theorem to find all solutions to the system of congruences $x \equiv 2 \pmod 3$, $x \equiv 1 \pmod 4$, and $x \equiv 3 \pmod 5$.

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$$x \equiv 2 \pmod 3$$
$$x \equiv 1 \pmod 4$$
$$x \equiv 3 \pmod 5$$

$m = 60$

$x = 2(20)y_1 + (1)(15)y_2 + (3)(12)y_3$

$20y_1 = 1 \pmod 3$

$20 = 6(3) + 2$

$3 = 2 + 1$

$1 = 3 - 2$

$= 3 - 20 + 6 \times 3$

$= 7 \times 3 - 20$

$y_1 \equiv -1 \pmod 3$

$\equiv (2)$

$15y_2 = 1 \pmod 4$

$15 = 3 \times 4 + 3$

$4 = 1 \times 3 + 1$

$1 = 4 - 3 \times 1$

$= 4 - 15 + 3 \times 4$

$= 4 \times 4 - 15$

$y_2 \equiv_4 -1 \equiv_4 3$

$12y_3 \equiv 1 \pmod 5$

$12 = 2 \times 5 + 2$

$5 = 2 \times 2 + 1$

$1 = 5 - 2 \times 2$

$= 5 - 2 \times [12 - 2 \times 5]$

$= 5 \times 5 - 2 \times 12$

$y_3 \equiv_5 -2 \equiv_5 3$

$X = 2(20) \times 2 + (1)(15)(3) + (3)(12)(3)$

$\equiv 233 \pmod{60}$

$\equiv 53$

Solve the system of congruence $x \equiv 3 \pmod 6$ and $x \equiv 4 \pmod 7$ using the method of back substitution.

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$$X \equiv 3 \pmod 6$$
$$X \equiv 4 \pmod 7$$

$X - 3 = 6n \Rightarrow X = 6n + 3$

$\Rightarrow 6n + 3 \equiv 4 \pmod 7$

$\Rightarrow 6n \equiv 1 \pmod 7$

$\Rightarrow n \equiv 6 \pmod 7 \Rightarrow n = 7m + 6$

$\Rightarrow X = 6(7m + 6) + 3$
$= 6 \times 7m + 36 + 3$
$= 42m + 39$

$X \equiv 39 \pmod{42}$

Find all solutions, if any, to the system of congruences $x \equiv 5 \pmod 6$, $x \equiv 3 \pmod{10}$, and $x \equiv 8 \pmod{15}$.

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$$X \equiv 5 \pmod 6$$
$$X \equiv 3 \pmod{10}$$
$$X \equiv 8 \pmod{15}$$

$\gcd(6, 10) = 2$ Thus, we cannot apply chinese reminder theorem.

However, using chinse reminder theorem, we can decode each equation

$$x \equiv 5 \pmod{6 = 2 \times 3}$$

$$\Downarrow$$

$$x \equiv 5 \equiv 1 \pmod 2$$
$$x \equiv 5 \equiv 2 \pmod 3$$

similarly,

$$x \equiv 3 \pmod{10 = 2 \times 5}$$

$$x \equiv 3 \equiv 1 \pmod 2 \checkmark$$
$$x \equiv 3 \equiv 3 \pmod 5$$

also, $x \equiv 8 \pmod{15 = 3 \times 5}$

$$x \equiv 8 \equiv 2 \pmod 3 \checkmark$$
$$x \equiv 8 \equiv 3 \pmod 5 \checkmark$$

$\Rightarrow$

$$x \equiv 1 \pmod 2$$
$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 5$$

$$x = (1)15\,y_1 + (2)(10)\,y_2 + (3)(6)\,y_3$$

$$15\,y_1 \equiv 1 \pmod 2$$
$$15 = 7 \times 2 + 1$$
$$\Rightarrow y_1 \equiv_2 1$$

$$10\,y_2 \equiv 1 \pmod 3$$
$$10 = 3 \times 3 + 1$$
$$y_2 \equiv_3 1$$

$$6\,y_3 \equiv 1 \pmod 5$$

$$y_3 = 1$$

$$\Rightarrow x = 15 + 20 + 18 = 53$$

$$X \equiv 53 \pmod{30}$$
$$\equiv 23 \quad \checkmark$$

Let $m_1, m_2, \ldots, m_n$ be pairwise relatively prime integers greater than or equal to 2. Show that if $a \equiv b \pmod{m_i}$ for $i = 1, 2, \ldots, n$, then $a \equiv b \pmod{m}$, where $m = m_1 m_2 \cdots m_n$. (This result will be used in Exercise 30 to prove the Chinese remainder theorem. Consequently, do not use the Chinese remainder theorem to prove it.)

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$$a \equiv b \pmod{m_1}$$
$$\vdots \qquad\qquad \vdots$$
$$a \equiv b \pmod{m_n}$$

$$\Rightarrow \quad \forall\, i, \quad m_i \mid a-b$$

Consider $m_1 \mid a-b$, $m_2 \mid a-b$ with $\gcd(m_1, m_2) = 1 \Rightarrow m_1, m_2$ has no common factors

$$\Rightarrow \quad a-b = m_1 n_1 = m_2 n_2$$
$$\Rightarrow \quad m_1 \mid n_2 \ \text{ & } \ m_2 \mid n_1$$

Thus, $a-b = m_1 m_2 k \qquad k \in \mathbb{Z}$

Thus, $a \equiv b \pmod{m_1 m_2}$

by induction,

$$a \equiv b \pmod{m = m_1 m_2 \cdots m_n}$$

Which integers are divisible by 5 but leave a remainder of 1 when divided by 3?

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$$x \equiv 0 \pmod 5$$
$$x \equiv 1 \pmod 3$$

$x = 5n \qquad n \in \mathbb{Z}$

$$5n \equiv 1 \pmod 3$$

$5 = 3 + 2$
$3 = 2 + 1$

$1 = 3 - 2$
$= 3 - 5 + 3$
$= (2)\times 3 - 5$

$n \equiv -1 \pmod 3$
$\equiv 2$

$\Rightarrow \quad n = 2 + 3k$

$x = (5)(3)k + 10$

$x \equiv 10 \pmod{15}$ ✓

Use Fermat's little theorem to find 23^{1002} **mod** 41.

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$\gcd(1002, 41) = \gcd(41, 18) = \gcd(18, 5)$
$= \gcd(5, 3) = \gcd(3, 2) = 1$

$\Rightarrow$ 41 is a prime

$a^{40} \equiv 1 \pmod{41}$ ←

$a^{41} \equiv a \pmod{41}$

$1002 = 40 \times 25 + 2$

$\Rightarrow \quad 23^{1002} \equiv_{41} (23^{40})^{25} \, 23^2 \equiv_{41} (1)^{25} \times 23^2$

$\equiv_{41} 529 \equiv_{41} 37$

Use Fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p.

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$a^{p-1} \equiv_p 1$

$\Rightarrow (a^{p-2})(a) \equiv_p 1 \Rightarrow a^{p-2}$ is the inverse a modulo p.

Use Fermat's little theorem to compute 3^{302} **mod** 5, 3^{302} **mod** 7, and 3^{302} **mod** 11.
Use your results from part (a) and the Chinese remainder theorem to find 3^{302} **mod** 385. (Note that $385 = 5 \cdot 7 \cdot 11$.)

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$a^{p-1} \equiv_p 1$

3^{302}, 5 $\quad p-1 = 4$

$302 = 75 \times 4 + 2$

$3^{302} \equiv_5 (3^4)^{75} 3^2 \equiv_5 3^2 \equiv_5 9 \equiv_5 4$

7, $\quad 7-1 = 6$

$302 = 6 \times 50 + 2$

$3^{302} \equiv_7 9 \equiv_7 2$

$11 - 1 = 10$

$302 = 30 \times 10 + 2$

$\Rightarrow \quad 3^{302} \equiv_{11} 9$

$\Rightarrow$

$$\begin{aligned} 3^{302} &\equiv 9 \pmod{11} \\ x \equiv 3^{302} &\equiv 2 \pmod{7} \\ 3^{302} &\equiv 4 \pmod{5} \end{aligned}$$

$x = 9(35)y_1 + 2(55)y_2 + 4(77)y_3$

$35y_1 \equiv 1 \mod 11$

$35 = 3 \times 11 + 2$

$11 = 5 \times 2 + 1$

$1 = 11 - 2 \times 5$

$= 11 - 5(35 - 3 \times 11)$

$\Rightarrow \quad \boxed{y_1 \equiv_{11} -5 \equiv_{11} 6}$

$55y_2 \equiv 1 \pmod 7$

$55 = 7 \times 7 + 6$

$1 = 7 - 6$

$= 7 - 55 + 7 \times 7$

$55 = 7 \times 7 + 6$
$7 = 6 + 1$

$1 = 7 - 6$
$= 7 - 55 + 7 \times 7$

$y_2 \equiv_7 -1 \equiv_7 6$

$77\, y_3 \equiv 1 \pmod 5$

$77 = 15 \times 5 + 2$
$5 = 2 \times 2 + 1$

$1 = 5 - 2 \times 2$
$= 5 - 2(77 - 15 \times 5)$

$y_2 \equiv_5 -2 \equiv_5 3$

$x = 9 \times 35 \times 6 + 2 \times 55 \times 6 + 4 \times 77 \times 3$
$\equiv 3474 \pmod{385}$
$\equiv 9$

Show with the help of Fermat's little theorem that if n is a positive integer, then 42 divides $n^7 - n$.

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$42 \mid n^7 - n$

$42 = 7 \times 6 = 7 \times 2 \times 3$

Thus, we only need to show that the prime factorization of $n^7 - n$ contains $7 \times 2 \times 3$ Thus,

$7 \mid n^7 - n$, $2 \mid n^7 - n$, $3 \mid n^7 - n$

$7 \mid n^7 - n$ is true by fermat's little theorem.

if n is odd $\Rightarrow n^7$ is odd $\Rightarrow n^7 - n$ is even
$\Rightarrow 2 \mid n^7 - n$. ✓

$3 \mid n^7 - n$?

$$n^7 \overset{?}{\equiv} n \pmod 3$$

using fermat's little theorem

$$n^2 \equiv 1 \pmod 3$$
$$n^3 \equiv n \pmod 3$$

$$7 = 2 \times 3 + 1$$

$$n^7 \equiv_3 (n^2)^3 n^1 \equiv_3 n$$ ✗

Thus, $42 \mid n^7 - n$